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Joint state and parameter estimation in quantum systems using cubature Kalman filtering^{☆,☆☆}Eram Taslima^{*}, Shyam Kamal, R.K. Saket

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ABSTRACT

This paper addresses the challenge of state estimation for two-level quantum systems governed by stochastic master equations, particularly when key Hamiltonian parameters are unknown. The critical parameters such as the qubit resonance frequency and the decay rate play a crucial role in determining system dynamics, hence their accurate estimation is essential for reliable state reconstruction. A robust framework based on the cubature Kalman filter (CKF) is developed that effectively handles both correlated and decorrelated noise processes inherent to quantum homodyne measurement. The proposed approach effectively mitigates performance degradation caused by parametric uncertainty, providing enhanced adaptability and robustness. Numerical simulations on a qubit in a cavity show that the CKF-based method achieves better estimation accuracy and faster convergence compared to the extended Kalman filter.

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1. Introduction

Quantum systems have become central to many modern technologies, including quantum computing (Yang et al., 2023), quantum communication (Vajner et al., 2022), and quantum sensing (Degen et al., 2017). These systems exploit quantum coherence and entanglement to perform tasks that are beyond the reach of classical devices (The ATLAS Collaboration, 2024). In quantum information processing, precise knowledge of the system state is essential for control, error correction, and reliable computation (Monroe, 2002). Similarly, in quantum metrology and communication, accurate estimation of quantum states directly determines measurement precision and information transfer efficiency (Giovannetti et al., 2011).

Therefore, accurate state estimation in quantum systems is a challenging task due to stochastic measurement backaction, system uncertainties, and environmental noise (Dong & Petersen, 2022). The dynamics of such systems are typically nonlinear and stochastic, driven by continuous weak measurements (Bouten

et al., 2007; Gao et al., 2019). In many practical scenarios, not only the system states but also certain model parameters, such as decoherence rates or resonant frequency, are unknown and time-varying. Hence, a joint estimation framework that can simultaneously identify both the quantum state and system parameters is essential to ensure robustness and adaptability in real-world implementations. Several estimation methods have been explored for quantum systems, including stochastic master equations (Bouten et al., 2007), Bayesian inference (Gammelmark & Mølmer, 2013), etc. However, these methods are often computationally intensive and not always suitable for real-time implementation with unknown parameters. Kalman filtering techniques (Särkkä & Svensson, 2023), on the other hand, provide a computationally efficient and recursive framework for optimal state estimation. For dynamical systems with nonlinear measurement and evolution dynamics, variants such as the extended Kalman filter (EKF) (Ribeiro, 2004), unscented Kalman filter (UKF) (Wan & Van Der Merwe, 2000), and cubature Kalman filter (CKF) (Arasaratnam & Haykin, 2009) have been developed to improve estimation accuracy. Among these, the CKF has shown strong performance in handling nonlinearities without requiring explicit Jacobian computation, making it well-suited for stochastic quantum state estimation.

The estimation of quantum states and Hamiltonian parameters remains a fundamental problem in feedback quantum control and quantum engineering. Classical results in quantum filtering, originating from Belavkin's quantum stochastic calculus (Belavkin, 1992) and formalized in Bouten et al. (2007), established the stochastic master equation (SME) as the foundation for continuous quantum measurement and estimation. Subsequent advances

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introduced quantum extensions of the Kalman filter (Emzir et al., 2017), incorporating measurement backaction and noise correlations inherent to homodyne detection. The authors in Rouchon and Ralph (2015) formulated efficient quantum filters suitable for feedback control of open systems. Building upon this theoretical groundwork, Corcione and Tarin (2023) developed an EKF framework for a driven and damped two-level system under continuous measurement, transforming the SME into an Itô state-space form and proving local observability of the Bloch dynamics. Their EKF simultaneously reconstructed the density matrix and identified the resonance and damping parameters, providing one of the first implementations of joint quantum state-parameter estimation. However, their procedure for parameter estimation is not clearly described, and several implementation aspects are not fully specified, making the method difficult to reproduce or extend for practical use. More recently, the authors in Dehaghani et al. (2024) advanced this direction by designing a decorrelated quantum EKF and a multiple-model adaptive estimator (MMAE) that account for correlated process and measurement noises, ensuring that estimated trajectories remain within the Bloch ball. Complementarily, Taslima et al. (2025) proposed an adaptive observer framework for state and parameter estimation of closed two-level systems using averaging theory.

While significant progress has been made in quantum state and parameter estimation through stochastic filtering and observer-based methods, most existing approaches rely on the EKF framework, which inherently suffers from linearization-induced inaccuracies when applied to nonlinear stochastic master equations. Moreover, adaptive observer formulations are often restricted to closed systems. Consequently, there remains a critical need for a nonlinear estimation framework that can jointly estimate the quantum state and unknown parameters with higher numerical stability and robustness to correlated noise. Therefore, this work proposes a robust CKF framework for stochastic quantum systems. The method ensures accurate quantum state reconstruction under process and measurement uncertainties and further extends to joint state and parameter estimation to enhance adaptability. To that end, the main contributions of this work are summarized as follows:

- (1) A CKF based state estimation framework is developed to achieve robust and accurate reconstruction of the quantum states under measurement noise and system uncertainties.
- (2) To further enhance robustness and adaptability, a joint CKF is proposed that simultaneously estimates both the system states and the unknown parameters, enabling effective state reconstruction and parameter identification.
- (3) Comparative analysis with the EKF demonstrates that the proposed CKF achieves significantly lower root-mean-square error (RMSE) in both state and parameter estimation, thereby confirming its estimation accuracy and robustness.

Notations. The set of real numbers is represented by $\mathbb{R}$, and $\mathbb{R}^n$ denotes the n -dimensional real vector space. $\mathbb{N}$ represents the set of natural numbers. The identity matrix of appropriate dimension is denoted by $\mathbf{I}$, and $\mathbf{0}$ represents a zero vector or matrix. The notation $[\cdot]^T$ indicates matrix or vector transpose, while $(\cdot)^\dagger$ denotes the Hermitian conjugate. The expectation operator is $\mathbb{E}[\cdot]$, and $\text{Tr}(\cdot)$ denotes the matrix trace. The set of all quantum density operators ρ defined on a finite-dimensional complex Hilbert space $\mathcal{H}$ is given by $\mathcal{H}_\rho = \{\rho \in \mathcal{L}(\mathcal{H}) \mid \rho = \rho^\dagger, \rho \geq 0, \text{Tr}(\rho) = 1\}$, where $\mathcal{L}(\mathcal{H})$ denotes the space of all linear operators acting on $\mathcal{H}$. The quantum expectation value is defined as $\langle X \rangle = \text{Tr}[X\rho]$. The operator $\|\mathbf{x}\|$ denotes the Euclidean norm of $\mathbf{x}$. The commutator of two operators A and B is $[A, B] = AB - BA$.

2. System description and problem formulation

2.1. Quantum stochastic dynamics

The evolution of a quantum system under continuous homodyne measurement can be described by a stochastic master equation (SME) in the Itô formalism. For a qubit system, the general form of the diffusive SME is given by:

$$d\rho(t) = -i[H_d + u(t)H_c, \rho(t)]dt + \Gamma \mathcal{D}[c_d]\rho(t)dt + M\mathcal{D}[c_m]\rho(t)dt + \sqrt{\eta M} \mathcal{H}[c_m]\rho(t)dW(t), \quad (1)$$

where $\rho \in \mathcal{H}_\rho$ denotes the density operator belonging to the space of physical density matrices. The unitary dynamics are governed by the system Hamiltonian, composed of a drift term H_d and a control term $u(t)H_c$, where $u(t)$ is the classical control input. The dissipative part of the dynamics is described by the Lindblad dissipation superoperators $\mathcal{D}[c]$ for the decoherence channel c_d and measurement channel c_m , defined for a generic operator c as:

$$\mathcal{D}[c]\rho = c\rho c^\dagger - \frac{1}{2}(c^\dagger c\rho + \rho c^\dagger c). \quad (2)$$

The measurement back-action is captured by the superoperator $\mathcal{H}[c]$:

$$\mathcal{H}[c]\rho = c\rho + \rho c^\dagger - \langle c + c^\dagger \rangle \rho. \quad (3)$$

The parameter $\Gamma > 0$ is the decay rate, $M > 0$ is the measurement strength, and $\eta \in [0, 1]$ is the detector efficiency. The stochastic increment $dW(t)$ is a Wiener process satisfying $\mathbb{E}[dW(t)] = 0$ and $\mathbb{E}[dW^2(t)] = dt$. The corresponding homodyne measurement current $J(t)$ is given by:

$$J(t)dt = \sqrt{\eta M} \langle c_m + c_m^\dagger \rangle dt + dW(t). \quad (4)$$

2.2. Coherence vector representation

For a single qubit, the density matrix can be parameterized using the coherence vector (or Bloch vector) $\mathbf{x}(t) = [\mathbf{x}_1(t), \mathbf{x}_2(t), \mathbf{x}_3(t)]^T \in \mathbb{R}^3$, which resides within the unit Bloch ball ($\|\mathbf{x}\| \leq 1$). The density matrix is expressed as:

$$\rho(t) = \frac{1}{2}\mathbf{I} + \frac{1}{2} \sum_{k=1}^3 \mathbf{x}_k(t)\sigma_k, \quad (5)$$

where σ_k are the Pauli matrices. Substituting (5) into the SME (1) yields the following Itô stochastic differential equation for the coherence vector:

$$d\mathbf{x}(t) = \mathbf{f}(\mathbf{x}, u(t))dt + \mathbf{g}(\mathbf{x})dW(t), \quad (6)$$

where the drift vector $\mathbf{f}(\mathbf{x}, \Omega) = [f_1, f_2, f_3]^T$ and the diffusion vector $\mathbf{g}(\mathbf{x}) = [g_1, g_2, g_3]^T$ are derived from the original SME and given as:

$$\begin{aligned} f_1(\mathbf{x}, \Omega) &= -\frac{\Gamma + M}{2}\mathbf{x}_1 - \omega_R\mathbf{x}_2 + 2u(t)\mathbf{x}_3, \\ f_2(\mathbf{x}, \Omega) &= \omega_R\mathbf{x}_1 - \frac{\Gamma + M}{2}\mathbf{x}_2, \\ f_3(\mathbf{x}, \Omega) &= -\Gamma(1 + \mathbf{x}_3) - 2u(t)\mathbf{x}_1, \end{aligned} \quad (7)$$

The diffusion terms, arising from the measurement back-action, are:

$$\begin{aligned} g_1(\mathbf{x}) &= \sqrt{\eta M}(1 - \mathbf{x}_3^2 - \mathbf{x}_1^2), \\ g_2(\mathbf{x}) &= -\sqrt{\eta M}\mathbf{x}_1\mathbf{x}_2, \\ g_3(\mathbf{x}) &= -\sqrt{\eta M}\mathbf{x}_1(1 + \mathbf{x}_3). \end{aligned} \quad (8)$$

The measurement current (4) can be rewritten as:

$$dy(t) = h(\mathbf{x}(t))dt + dV(t), \quad (9)$$

where $dy(t)$ is the observed measurement increment, $h(\mathbf{x}) = \sqrt{\eta M} \mathbf{x}_3$ is the measurement function, and $dV(t)$ is a noise increment that incorporates both the fundamental quantum noise $dW(t)$ and any additional classical noise $dZ(t)$ from the measurement apparatus, satisfying $\mathbb{E}[dV^2(t)] = \sigma^2 dt$.

2.3. Problem statement

Given the quantum system dynamics described by (6) and the continuous measurement stream (9), develop a set of estimators to accurately track the evolution of the coherence vector $\mathbf{x}(t)$ (and hence the density operator $\rho(t)$). A key challenge is to maintain estimation performance when the model contains unknown, constant parameters, such as the decay rate Γ and resonant frequency ω_R .

3. Cubature Kalman filtering for quantum systems

The CKF is a powerful nonlinear estimation technique that provides a more accurate and numerically stable alternative to the EKF for systems with nonlinear dynamics and Gaussian noise. It is based on the principle of cubature rules to approximate multidimensional integrals of the form $\int_{\mathbb{R}^n} f(\mathbf{x}) \exp(-\mathbf{x}^T \mathbf{x}) d\mathbf{x}$, which commonly appear in the Bayesian filtering equations. Unlike the EKF, which relies on first-order linearization and may suffer from bias or divergence in strongly nonlinear regimes, the CKF approximates Gaussian-weighted integrals using cubature rules, yielding higher-order accuracy in moment propagation. This property is particularly important for quantum systems, where the dynamics are nonlinear and constrained to the Bloch ball. As a result, CKF demonstrates improved numerical stability and robustness compared to EKF, especially near the physical boundary of the state space.

3.1. CKF with correlated noise

The first step is to ensure observability of the qubit system using the following Lemma.

Lemma 1 (Local Observability Condition [Isidori, 1995](#)). Consider a nonlinear system

$$\dot{\mathbf{x}} = f(t, \mathbf{x}), \quad \mathbf{y} = h(\mathbf{x}), \quad (10)$$

with $\mathbf{x}(0) = \mathbf{x}_0 \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}$. The system is said to be locally observable if the observability matrix $\mathcal{O}(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} [L_f^0 h(\mathbf{x}) \quad L_f^1 h(\mathbf{x}) \quad \cdots \quad L_f^{n-1} h(\mathbf{x})]^T$ satisfies $\text{rank } \mathcal{O}(\mathbf{x}) = n$, where $L_f h(\mathbf{x}) = \frac{\partial h}{\partial \mathbf{x}} f(t, \mathbf{x})$ denotes the Lie derivative of h along f .

For the stochastic Bloch dynamics (6) and (9), this condition is fulfilled for all $t \neq \frac{\pi(2n+1)}{2\omega_F}$, $n \in \mathbb{N}$ when driven with sinusoidal input $\Omega_F \cos(\omega_F t)$. Hence, the full Bloch vector $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3]^T$ and equivalently the density operator ρ can be reconstructed from the continuous measurement record (9). The condition $t \neq \frac{\pi(2n+1)}{2\omega_F}$ arises from this particular measurement-control configuration, and should not be interpreted as a universal observability condition for general quantum systems or alternative measurement channels.

The discrete-time equivalent of the continuous-time system (6) and (9) at time step k can be written as follows by applying the Euler–Maruyama discretization with time step Δt as follows:

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, u_{k-1}) + \mathbf{w}_k, \quad (11)$$

$$y_k = h(\mathbf{x}_k) + v_k, \quad (12)$$

where $f(\mathbf{x}_{k-1}, u_{k-1}) = \mathbf{x}_{k-1} + f(\mathbf{x}_{k-1}, u_{k-1})\Delta t$ and $h(\mathbf{x}_k) = h(\mathbf{x}_k)\Delta t$. Also $\mathbf{w}_k = g(\mathbf{x}_{k-1})\Delta W_{k-1}$ and $v_k = \Delta V_k$ are discrete noise processes with covariance matrices $\mathbf{Q}_k$ and R_k respectively, and cross-covariance $\mathbf{C}_k = \mathbb{E}[\mathbf{w}_k v_k^T]$.

In the quantum filtering problem, a key challenge is the correlation between the process noise $\mathbf{w}_k$ in the state equation and the measurement noise v_k in the output equation. Specifically, we have $\mathbb{E}[\mathbf{w}_k v_k^T] = \mathbb{E}[g(\mathbf{x}_{k-1})\Delta W_k(\Delta W_k + \Delta Z_k)] = g(\mathbf{x}_{k-1})\Delta t \neq 0$, since $\mathbb{E}[\Delta W_k^2] = \Delta t$ and $\mathbb{E}[\Delta W_k \Delta Z_k] = 0$. This correlation must be explicitly accounted for in the filter design to avoid performance degradation. For a system with correlated noises, the standard CKF prediction and update steps are modified using framework of [Ge et al. \(2014\)](#) as given in Algorithm 1.

Remark 1. In practice, $\mathbf{Q}_k$ can be approximated as only a constant matrix $\mathbf{Q}$ representing the average diffusion level, particularly when $g(\mathbf{x})$ varies slowly or the time step Δt is small. This simplification maintains good performance while reducing computational complexity. Our simulations show robustness to this approximation, with constant $\mathbf{Q}$ yielding similar RMSE to the state-dependent computation as shown in [Table 2](#).

3.2. Noise decorrelation and the CKF

To circumvent the complexity of handling correlated noise directly, we can also employ the noise decorrelation approach ([Dehaghani et al., 2024](#)).

Proposition 1 (Noise Decorrelation). Consider the stochastic differential equation for the coherence vector (6) and the measurement Eq. (9). Define a new process $\tilde{\mathbf{x}}(t)$ and a new Wiener process $d\tilde{W}(t)$ as follows:

$$d\tilde{\mathbf{x}}(t) = f(\tilde{\mathbf{x}}, u(t))dt - \bar{\sigma} g(\tilde{\mathbf{x}})h(\tilde{\mathbf{x}})dt + \bar{\sigma} g(\tilde{\mathbf{x}})dy(t), \quad (13)$$

$$d\tilde{W}(t) = dW(t) - \bar{\sigma} dV(t), \quad (14)$$

where $\bar{\sigma} = 1/(1 + \sigma^2)$. Then, the process noise $d\tilde{W}(t)$ and measurement noise $dV(t)$ in the transformed system are decorrelated, i.e., $\mathbb{E}[d\tilde{W}(t)dV(t)] = 0$. Moreover, the transformed system is equivalent to the original one in the sense that $\mathbf{x}(t) = \tilde{\mathbf{x}}(t)$ for all $t \geq t_0$ given the same initial conditions and exogenous signals.

With this transformation, we now have a system with decorrelated noises. The discrete-time version of the transformed state Eq. (13) becomes:

$$\tilde{\mathbf{x}}_k = \tilde{\mathbf{f}}_{k-1}(\tilde{\mathbf{x}}_{k-1}, u_{k-1}, y_{k-1}) + \tilde{\mathbf{w}}_{k-1}, \quad (15)$$

where $\tilde{\mathbf{w}}_k$ is the discrete process noise with covariance $\tilde{\mathbf{Q}}_k$, and it is decorrelated with v_k . The standard CKF algorithm can now be applied directly to this transformed system.

The CKF steps for the decorrelated system are identical to the CKF algorithm 1, but using the transformed dynamics $\tilde{\mathbf{f}}_{k-1}$ and noting that the cross-covariance $\mathbf{C}_k = 0$, thus bypassing steps 10 and 11. This significantly simplifies the filter implementation while maintaining optimal estimation performance.

3.3. Filter projection

To ensure that the estimated coherence vector $\hat{\mathbf{x}}_{k|k}$ always represents a physically valid quantum state (i.e., it remains within the Bloch ball, $\|\hat{\mathbf{x}}_{k|k}\| \leq 1$), we apply a projection step after the measurement update:

$$\hat{\mathbf{x}}_{k|k} \leftarrow \text{Proj}(\hat{\mathbf{x}}_{k|k}) = \begin{cases} \hat{\mathbf{x}}_{k|k}, & \text{if } \|\hat{\mathbf{x}}_{k|k}\| \leq 1 \\ \hat{\mathbf{x}}_{k|k} / \|\hat{\mathbf{x}}_{k|k}\|, & \text{otherwise.} \end{cases} \quad (16)$$

Algorithm 1 CKF Algorithm for State Estimation

Require: Dimension $n = 3$, Prior estimate $\hat{\mathbf{x}}_{k-1|k-1}$, error covariance $\mathbf{P}_{k-1|k-1}$, measurement noise covariance $\mathbf{R}_k$, nonlinear mappings $\mathbf{f}_k(\cdot)$, $\mathbf{h}_k(\cdot)$, diffusion term $\mathbf{g}(\cdot)$, time step Δt , $\mathbf{Q}_c$ (optional).

Ensure: Updated state $\hat{\mathbf{x}}_{k|k}$ and covariance $\mathbf{P}_{k|k}$.

Prediction:

1: Factorize $\mathbf{P}_{k-1|k-1} = \mathbf{S}_{k-1|k-1} \mathbf{S}_{k-1|k-1}^\top$ and form cubature points

$$\mathbf{X}_{i,k-1|k-1} = \hat{\mathbf{x}}_{k-1|k-1} + \mathbf{S}_{k-1|k-1} \boldsymbol{\xi}_i,$$

where $\boldsymbol{\xi}_i = \sqrt{n}[\mathbf{1}]_i$ for $i = 1, \dots, 2n$ and $[\mathbf{1}]_i$ is the i -th column of $[\mathbf{I}_n - \mathbf{I}_n]$.

2: Propagate each point through system dynamics:

$$\mathbf{X}_{i,k|k-1}^* = \mathbf{f}_{k-1}(\mathbf{X}_{i,k-1|k-1}).$$

3: Compute predicted mean:

$$\hat{\mathbf{x}}_{k|k-1} = \frac{1}{2n} \sum_{i=1}^{2n} \mathbf{X}_{i,k|k-1}^*.$$

4: Compute process noise covariance from cubature points:

$$\mathbf{Q}_k = \frac{1}{2n} \sum_{i=1}^{2n} \mathbf{g}_{k-1}(\mathbf{X}_{i,k-1|k-1}) \mathbf{g}_{k-1}(\mathbf{X}_{i,k-1|k-1})^\top \Delta t + \mathbf{Q}_c.$$

5: Compute predicted covariance:

$$\mathbf{P}_{k|k-1} = \frac{1}{2n} \sum_{i=1}^{2n} (\mathbf{X}_{i,k|k-1}^* - \hat{\mathbf{x}}_{k|k-1})(\mathbf{X}_{i,k|k-1}^* - \hat{\mathbf{x}}_{k|k-1})^\top + \mathbf{Q}_k.$$

Update:

6: Factorize $\mathbf{P}_{k|k-1} = \mathbf{S}_{k|k-1} \mathbf{S}_{k|k-1}^\top$ and set

$$\mathbf{X}_{i,k|k-1} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{S}_{k|k-1} \boldsymbol{\xi}_i.$$

7: Compute cross-covariance at predicted state:

$$\mathbf{C}_k = \mathbf{g}(\hat{\mathbf{x}}_{k-1|k-1}) \Delta t.$$

8: Propagate through measurement model:

$$\mathbf{Y}_{i,k|k-1} = \mathbf{h}_k(\mathbf{X}_{i,k|k-1}).$$

9: Compute predicted measurement and the CKF measurement covariances:

$$\hat{\mathbf{y}}_{k|k-1} = \frac{1}{2n} \sum_{i=1}^{2n} \mathbf{Y}_{i,k|k-1},$$

$$\mathbf{P}_{yy}^{\text{CKF}} = \frac{1}{2n} \sum_{i=1}^{2n} (\mathbf{Y}_{i,k|k-1} - \hat{\mathbf{y}}_{k|k-1})(\mathbf{Y}_{i,k|k-1} - \hat{\mathbf{y}}_{k|k-1})^\top + \mathbf{R}_k,$$

$$\mathbf{P}_{xy}^{\text{CKF}} = \frac{1}{2n} \sum_{i=1}^{2n} (\mathbf{X}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1})(\mathbf{Y}_{i,k|k-1} - \hat{\mathbf{y}}_{k|k-1})^\top.$$

10: Compute $\mathbf{J}_k = \left. \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}_{k|k-1}}$.

11: Correlated-noise corrections:

$$\mathbf{P}_{xy} = \mathbf{P}_{xy}^{\text{CKF}} + \mathbf{C}_k,$$

$$\mathbf{P}_{yy} = \mathbf{P}_{yy}^{\text{CKF}} + \mathbf{J}_k \mathbf{C}_k + \mathbf{C}_k^\top \mathbf{J}_k^\top.$$

12: Update Kalman gain, state, and error covariance:

$$\mathbf{K}_k = \mathbf{P}_{xy} \mathbf{P}_{yy}^{-1},$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1}),$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{P}_{yy} \mathbf{K}_k^\top.$$

This projection guarantees that the reconstructed density matrix $\hat{\rho}_k$ from (5) satisfies all physical properties of a quantum state.

The proposed CKF framework is, in principle, extensible to multi-qubit and higher-dimensional quantum systems. The number of cubature points grows linearly with the state dimension ($2n$), while the dimension of the generalized Bloch vector increases as $d^2 - 1$ for a d -level system, leading to a rapid growth of the effective state space. Moreover, enforcing physical constraints such as positive semidefiniteness of the estimated density matrix becomes increasingly complex as the system dimension increases. Consequently, although the algorithmic structure of the CKF scales systematically, practical implementation beyond a few qubits remains challenging due to the combined effects of high-dimensional state spaces and strong nonlinearities inherent in quantum dynamics.

4. Parameter identification with CKF

4.1. Single parameter unknown

Consider the case where the amplitude damping rate $\Gamma > 0$ is unknown and constant. The measured homodyne current remains (9). To ensure (local) identifiability of Γ from the measurement $y(t)$, the input $u(t)$ must provide persistent excitation (PE) of the Bloch dynamics.

Lemma 2. Consider the driven, damped two-level quantum system with σ_y control

$$\begin{aligned}\dot{\mathbf{x}}_1 &= -\frac{\Gamma+M}{2}\mathbf{x}_1 - \omega_R\mathbf{x}_2 + 2\Omega_F \cos(\omega_F t)\mathbf{x}_3, \\ \dot{\mathbf{x}}_2 &= \omega_R\mathbf{x}_1 - \frac{\Gamma+M}{2}\mathbf{x}_2, \\ \dot{\mathbf{x}}_3 &= -\Gamma(1+\mathbf{x}_3) - 2\Omega_F \cos(\omega_F t)\mathbf{x}_1, \quad y = \sqrt{\eta M}\mathbf{x}_3,\end{aligned}$$

where Γ is an unknown but constant parameter. Define the augmented state $z = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \Gamma]^\top$ with $\dot{\Gamma} = 0$. Then, for all $(\mathbf{x}, t)$ such that

$$\Omega \neq 0, \quad \omega_R \neq 0, \quad \Gamma \neq 0, \quad \cos(\omega_F t) \neq 0, \quad (17)$$

the extended observability matrix

$$\mathcal{O}(z, t) = \frac{\partial}{\partial z} [h \quad L_f h \quad L_f^2 h \quad L_f^3 h]^\top$$

has full rank 4, and therefore Γ is locally identifiable. These identifiability conditions are model-dependent and rely explicitly on the chosen σ_z measurement, σ_y control structure, and sinusoidal excitation.

Proof. The output is $h = \sqrt{\eta M}\mathbf{x}_3$, hence $\partial h / \partial z = [0 \ 0 \ \sqrt{\eta M} \ 0]$, which provides the $\mathbf{x}_3$ -direction. The first Lie derivative along the augmented dynamics is

$$L_f h = \sqrt{\eta M}[-\Gamma(1+\mathbf{x}_3) - 2\Omega_F \cos(\omega_F t)\mathbf{x}_1],$$

so its gradient is $\frac{\partial(L_f h)}{\partial z} = [-2\sqrt{\eta M}\Omega_F \cos(\omega_F t) \ 0 \ -\sqrt{\eta M}\Gamma \ -\sqrt{\eta M}(1+\mathbf{x}_3)]$. This row contains simultaneously (i) the $\mathbf{x}_1$ -direction through the control term $-2\Omega_F \cos(\omega_F t)\mathbf{x}_1$, and (ii) the parameter direction through $-\Gamma(1+\mathbf{x}_3)$. Hence, under (17), the first two rows of $\mathcal{O}$ already span $\{\mathbf{x}_1, \mathbf{x}_3, \Gamma\}$. The second Lie derivative $L_f^2 h$ depends on $\dot{\mathbf{x}}_1$ and $\dot{\mathbf{x}}_2$. Since $\dot{\mathbf{x}}_2 = \omega_R\mathbf{x}_1 - \frac{\Gamma+M}{2}\mathbf{x}_2$ and $\dot{\mathbf{x}}_1$ itself contains the driven term $2\Omega_F \cos(\omega_F t)\mathbf{x}_3$, the gradient $\partial(L_f^2 h)/\partial z$ produces a term proportional to $\Omega_F \cos(\omega_F t)\omega_R$ in the $\mathbf{x}_2$ -component. Therefore, for $\Omega \neq 0$, $\omega_R \neq 0$, and $\cos(\omega_F t) \neq 0$, the third row brings in the $\mathbf{x}_2$ -direction. Finally, $L_f^3 h$ removes possible degeneracies at isolated $(\mathbf{x}, t)$ and preserves the full span. Thus the four rows

$$\frac{\partial h}{\partial z}, \quad \frac{\partial(L_f h)}{\partial z}, \quad \frac{\partial(L_f^2 h)}{\partial z}, \quad \frac{\partial(L_f^3 h)}{\partial z}$$

are linearly independent for generic $(\mathbf{x}, t)$ satisfying (17), so $\text{rank } \mathcal{O}(z, t) = 4$. By the standard nonlinear observability test (e.g. [Isidori \(1995\)](#)), the augmented system is locally observable, and since $\dot{\Gamma} = 0$, the constant parameter Γ is locally identifiable. $\square$

If the control field $u(t)$ were constant or zero, the system would relax monotonically to the ground state $\mathbf{x}_3 = -1$, causing the homodyne output to saturate and making Γ unidentifiable. Hence, a persistently exciting (PE) modulation is needed. Then, for joint state and parameter estimation, we augment the state vector of Algorithm 1 with unknown parameters and set their dynamics to zero.

4.2. Estimation of two parameters with CKF

In practical scenarios, both the amplitude damping rate Γ and the frequency ω_R may be uncertain. To address this, the CKF can be extended to jointly estimate the three system states $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ and the two constant parameters (Γ, ω_R) . The augmented state vector is defined as

$$z = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \Gamma, \omega_R]^\top, \quad \dot{\Gamma} = 0, \quad \dot{\omega}_R = 0,$$

The CKF algorithm follows the same prediction-update structure described in Algorithm 1, with the process model $\dot{z} = f(t, z) + \xi(t)$ and the measurement $y = \sqrt{\eta M}\mathbf{x}_3 + W_t$, where $\xi(t)$ is the process noise representing modeling uncertainty in both the physical and parameter subspaces.

Further, applying the nonlinear observability lemma ([Isidori, 1995](#)) to the augmented dynamics shows that the rank of the extended observability matrix

$$\mathcal{O}(z, t) = \frac{\partial}{\partial z} [h \quad L_f h \quad L_f^2 h \quad L_f^3 h \quad L_f^4 h]^\top$$

is generically full, i.e. $\text{rank } \mathcal{O}(z, t) = 5$ for

$$\Omega \neq 0, \quad \Gamma \neq 0, \quad \omega_R \neq 0, \quad \cos(\omega_F t) \neq 0.$$

Hence, all five variables $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \Gamma, \omega_R)$ are locally observable. The parameter Γ appears in the dissipative terms of both the transverse and longitudinal channels, while ω_R enters the rotational coupling between $\mathbf{x}_1$ and $\mathbf{x}_2$. Consequently, in general, a control field that excites both coherence and population dynamics is necessary to observe these parameters.

Remark 2. The noise decorrelation transform of Section 3.2 is statistically equivalent to the original model at the level of the exact nonlinear filter, hence no information is lost in principle. However, when one uses approximate Gaussian filters, such as approximate Q, R, and cubature quadrature the transform can inadvertently attenuate the parameter sensitivity carried by the state measurement noise cross-covariance $S_k = \mathbb{E}[w_k v_k^\top]$. In particular, explicitly modeling the correlation can yield larger innovation gains for the parameter. We observe that joint state parameter CKF with the correlated-noise formulation recovers unknown parameters more reliably.

5. Result and discussion

We evaluate the performance of the proposed CKF and compare it with the standard EKF for different scenarios. For the analysis, we consider the qubit following (6)–(8) with observation (9). In density operator framework, this can be given using the system Hamiltonian as

$$H(t) = \frac{\omega_R}{2}\sigma_z + u(t)\frac{\sigma_y}{2},$$

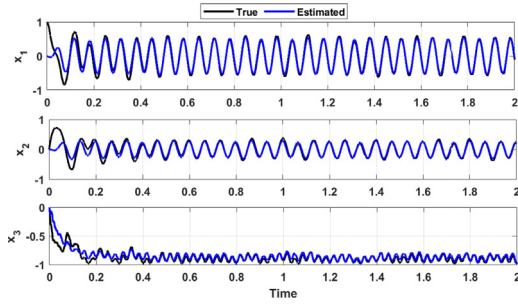


Fig. 1. State estimation of qubit using CKF (without decorrelation).

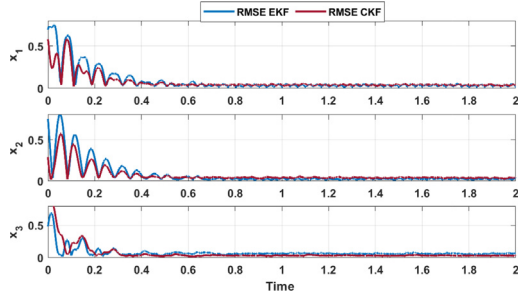


Fig. 2. Average RMSE of 50 realizations with CKF and EKF with correlated noise.

where ω_R is the Rabi frequency and $u(t)$ is the control amplitude. The dissipation operator is $c_d = \sqrt{\Gamma/2} \sigma_z$, corresponding to dephasing along the $\mathbf{x}_3$ -axis, and the measurement operator is $c_m = \sqrt{M/2} \sigma_z$. The associated measurement current follows $dy(t) = \sqrt{\eta M} \mathbf{x}_3(t) dt + dV(t)$. All simulation studies in this section strictly follow the modeling assumptions used in Lemma 2.

5.1. State estimation

This section represents the CKF implementation, where the filter operates under correlated and decorrelated process and measurement noises arising from the common Wiener increment in the quantum stochastic master equation. The true system parameters are chosen as $\Gamma = 10$, $\omega_R = 50$, measurement strength $M = 0.6$, and detection efficiency $\eta = 0.8$. The control signal is applied along the $\mathbf{x}_2$ -axis as $u(t) = 40 \cos(30\pi t)$. The noise covariances used in the CKF are $\mathbf{Q}_c = \text{diag}(10^{-6}, 10^{-6}, 10^{-5})$ and $R = \frac{\sigma^2}{dt} = 100$. The initial true state is $\mathbf{x}_0 = [1, 0, 0]^T$, and the initial estimate is $\hat{\mathbf{x}}_0 = [0, 0, 0]^T$ with error covariance $P_0 = \text{diag}(1, 1, 1)$. Fig. 1 shows the true and estimated Bloch states trajectory evolutions under correlated noise. To assess the performance, we run the filters (CKF and EKF) for 50 Monte-Carlo trials and the corresponding RMSE is shown in Fig. 2. Similarly, for the case of decorrelated noise, the results are shown in Fig. 3 (state estimation) and Fig. 4 (RMSE). Furthermore, to quantify the estimation performance, Table 1 shows the time-averaged RMSE of the 50 realizations with both the EKF and CKF. It can be verified that the CKF achieves a lower RMSE value and is therefore more robust than the EKF. For comparison, we also evaluate performance using a constant process noise covariance $\mathbf{Q}$. The results in Table 2 confirm that a suitably chosen constant $\mathbf{Q}$ provides performance comparable to the state-dependent approach.

5.2. Joint estimation of Γ

We now consider jointly estimating the unknown constant parameter Γ along with the Bloch states. For the joint estimation

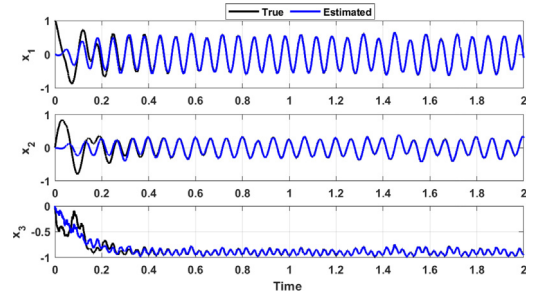


Fig. 3. State estimation of qubit using CKF (with decorrelation).

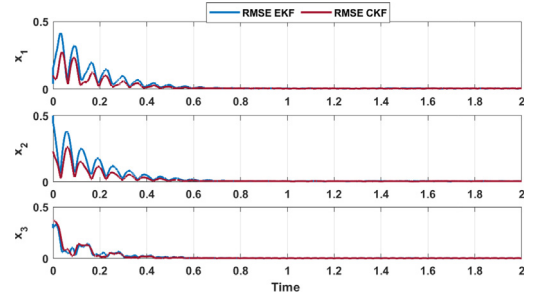


Fig. 4. Average RMSE of 50 realizations with CKF and EKF with decorrelated noise.

Table 1

Estimation performance under decorrelated and correlated innovations.

State	Decorrelated		Correlated	
	CKF	EKF	CKF	EKF
x_1	0.0231	0.0332	0.0710	0.0836
x_2	0.0216	0.0346	0.0686	0.0756
x_3	0.0154	0.0182	0.0657	0.0786

Table 2

Sensitivity of CKF performance to constant $\mathbf{Q} = q\mathbf{I}_3$.

q	x_1 RMSE	x_2 RMSE	x_3 RMSE
10^{-8}	0.0847	0.0896	0.0634
10^{-7}	0.0828	0.0870	0.0634
10^{-6}	0.0862	0.0911	0.0658
10^{-5}	0.0824	0.0862	0.0631
10^{-4}	0.0622	0.0560	0.0978
10^{-3}	0.3224	0.2563	0.6942

of Γ , the true initial state is $\mathbf{x}_0 = [0.5, 0.5, 0.5]^T$, while the filter is initialized with $\hat{\mathbf{x}}_0 = [0, 0, 0, 5]^T$. The true and estimated Bloch states are shown in Fig. 5 and the parameter estimate is shown in Fig. 6. The result verifies the estimation of the unknown parameter and states within 1 time unit, hence establishing a fast and robust approach. We validated the robustness of the correlated framework through 10 simulation runs, using Algorithm 1 (correlated) and compared with the decorrelated method from Proposition 1. The results, presented in Fig. 7, demonstrate that the correlated algorithm provides reliable estimates, its mean across runs converges accurately to the true parameter, whereas the decorrelated estimator shows noticeable deviation and higher variability.

5.3. Estimating Γ and ω_R

We now consider the case where both the parameters Γ and ω_R are unknown. For better identifiability, we consider the control $u(t) = A + \Omega_F \cos(\omega_F t)$, where $A = -20$, $\Omega = 25$ and

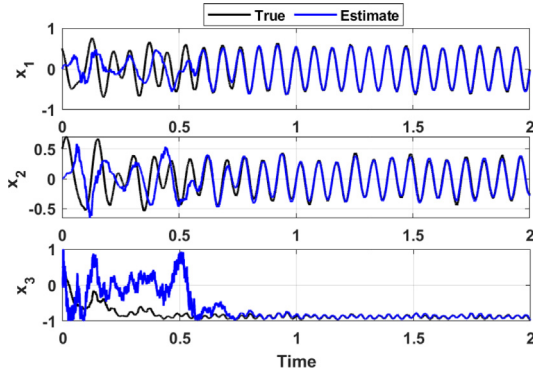


Fig. 5. Estimation of Bloch states with CKF when Γ is unknown.

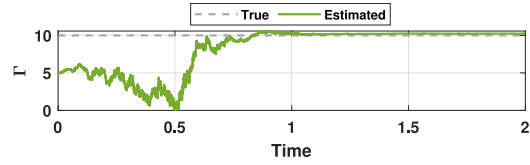


Fig. 6. Estimation of unknown parameter Γ using joint state and parameter CKF.

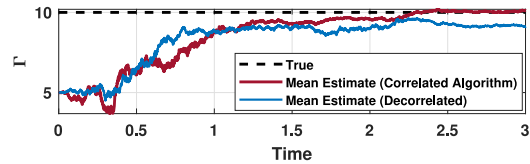


Fig. 7. Mean estimate of the unknown parameter Γ obtained using joint state-parameter CKF under correlated and decorrelated noise formulations, averaged over 10 Monte Carlo realizations.

$\omega_F = 200\pi$. Figs. 8 and 9 shows the true and estimated state with CKF and EKF when both the parameters were unknown. Fig. 10 shows the parameter estimation with CKF and EKF. These results verify the more accurate and robust estimation of states and parameters with the CKF than with the EKF. The more dynamic appearance of the trajectories stems from the joint estimation of Γ and ω_R under a high-frequency, persistently exciting control signal.

6. Conclusion

This paper developed a CKF framework for continuously measured qubit systems that handles the two main practical challenges in quantum estimation: nonlinear, noise-correlated stochastic dynamics and unknown Hamiltonian/dissipation parameters. By starting from the SME and working in the Bloch-vector form, we demonstrated how to treat process-measurement noise correlation explicitly (or eliminate it via a decorrelation transform) so that a CKF can be applied without sacrificing the physicality of the quantum state, which is enforced through a Bloch-ball projection. We then augmented the state with unknown but constant parameters (notably the damping rate Γ and the resonance frequency ω_R) and used nonlinear observability arguments under persistently exciting control to justify joint estimation. Simulations on a driven, damped qubit under homodyne detection demonstrated that, across both correlated and decorrelated cases, the CKF converges faster and attains lower RMSE than the EKF for state reconstruction, and that the joint CKF successfully identifies the unknown parameters

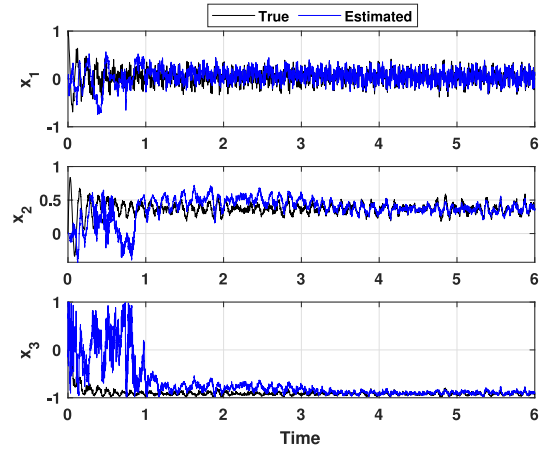


Fig. 8. True and estimated Bloch states using CKF when both Γ and ω_R are unknown.

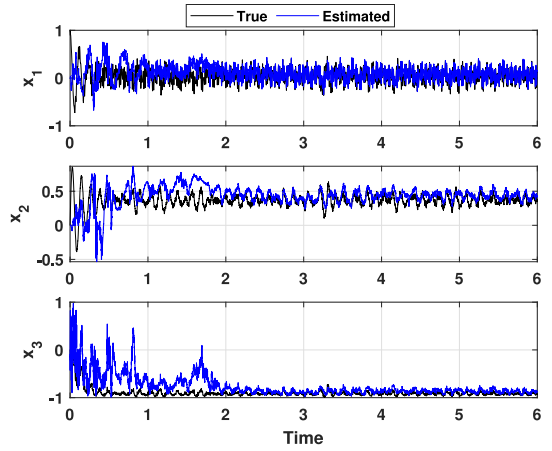


Fig. 9. True and estimated Bloch states using EKF when both Γ and ω_R are unknown.

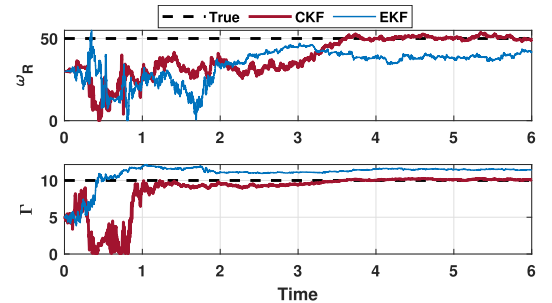


Fig. 10. Parameter estimation using EKF and CKF.

while retaining accurate state tracking, establishing CKF as a more robust alternative to EKF for real-time quantum filtering with parametric uncertainty.

CRediT authorship contribution statement

Eram Taslima: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Shyam Kamal:** Validation, Supervision, Methodology, Conceptualization. **R.K. Saket:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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